

ALL SKELETONS ARE CREATED EQUAL! A DOMAIN ADAPTATION TRANSFORMER TO HANDLE MULTIPLE TOPOLOGIES

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ABSTRACT

Designing an effective Human Pose Estimation (HPE) pipeline necessitates handling diverse datasets, a task often deemed necessary but burdensome by researchers. Existing datasets are annotated using different conventions, with the human body represented as a parametric 3D model or as a collection of 2D/3D joints and bones, known as skeleton-based annotation. Despite its widespread use in training both 2D and 3D HPE networks, the lack of standardization in the topologies of joint-based pose annotations requires considerable difficulties when evaluating the algorithms performances across different datasets. To solve this issue, we introduce a novel self-supervised human pose domain adaptation approach to map a given skeletal model into a target model of choice. We design a transformer-based architecture trained to reconstruct missing joints within a given topology, aligning them with the target model. During testing, the network seamlessly reconstructs the missing joints, treating them as if masked, based on the common joints shared by the original topology and the target one. Unlike previous approaches, our method works with arbitrary body poses, joints number, and skeleton topologies, across multiple datasets.¹

Index Terms— Skeletons, Poses, Topologies

1. INTRODUCTION

Human pose estimation (HPE) datasets [3,4] represent the human skeleton with kinematic models consisting of a varying number of 15-30 joints (as reported in Table 1).

Having different skeletal models to represent the human pose makes it difficult for researchers to test and compare methods across different datasets [3]. The ambiguity between different skeletal models comes in a variety of fashions: position of the joint (e.g. the nose as the center of the face or as the point between the 2 eyes), identity of the joint (e.g. MidHip or pelvis) or position of the annotation on the skin surface (e.g., the shoulder can be annotated at the top, under the armpit, in the frontal part or in the back). In addition, when projected on

¹The code is available at <https://github.com/mmlab-cv/DAT.git>

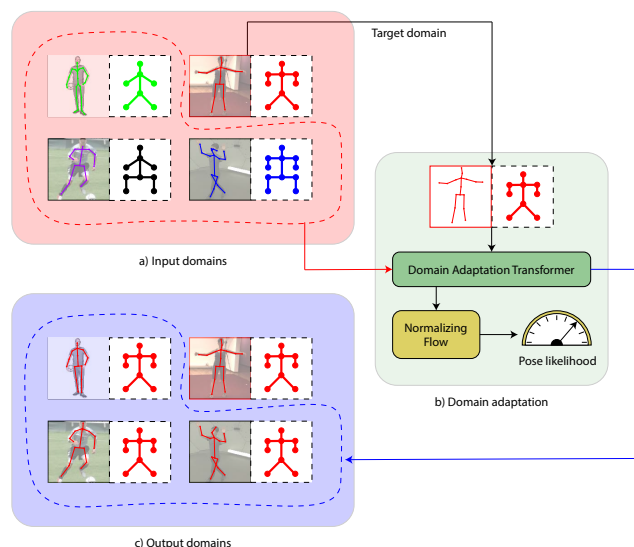


Fig. 1: (a) Various Human Pose Estimation (HPE) datasets, representing different *input domains*, utilize distinct skeleton models. These models can differ in joints number, joints position, topology, and shape characteristics (e.g., flat or angled shoulders). (b) Upon selecting a *target domain*, we employ a *Domain Adaptation Transformer* (DAT) [1] to transform the input skeleton from any given dataset format into the chosen target format. A *Normalizing Flow* (NF) [2] is used to assess the likelihood of a generated pose to be realistic. (c) Next, all *output domains* are annotated in the format of the chosen *target domain* for standardization purposes.

a 2D image, the complexity of the human body shape almost inevitably leads to occluded joints, which position has to be *guessed* when annotating. As reported in Table 1, the relevant datasets in the literature display different annotation formats. Although they are all skeleton-based, they do not comply to a standard pose model [5,6].

To overcome these non-trivial limitations, one can train the same network on different topologies, or implement different decoding heads in the architecture, one for each topology. However, learning different topologies turns into an inef-

efficient allocation of resources, in terms of both computational burden and network size, especially at training time [6, 7]. A sub-optimal solution is to uniform different skeleton models by using manually-defined rules to add/remove joints when an input skeleton has fewer/more joints than the target one.

If a target skeleton model has **fewer** joints than the source one (e.g. going from SMPL [8] to H3.6M [9]), dropping the extra joints might not work. Although the remaining joints (e.g. hips) could find a match in the target skeleton, their position can be inconsistent (see Fig. 2); aligning two models still requires a transformation in the 3D space for each pair of source-target joints.

If a target skeleton has **more** joints than the source one, interpolation is a viable option (e.g. adding a joint along the spine). However, issues might occur at the leaf joints (e.g. hands and feet), as there would be no awareness on the global pose prior.

In this work, we show how to seamlessly adapt the skeleton model to target different topologies. We propose a network that can perform unsupervised domain adaptation between different pose models, allowing for the harmonization of the different templates towards an arbitrarily chosen model, as shown in Figure 1. Recent approaches [10, 11] demonstrate how training an architecture to reconstruct the missing parts of a masked input is an effective strategy to obtain a rich representation of the input data. Leveraging the same principles, we aim at reconstructing the complete version of a masked skeleton, where the masked joints are the ones missing in the original topology. As shown in Figure 3, we design a transformer-based [1, 12] architecture that allows to upscale the annotation of the skeletons to the target topology.

To the best of our knowledge, our work is the first that explicitly aims at performing the self-supervised human pose domain adaptation task, and applying the masking strategy to pose modeling.

The main contribution of our work can be summarised as:

- we propose a novel self-supervised pipeline to uniform skeleton representation between datasets, without relying on manual mapping or complex generative models.
- our approach seamlessly applies to existing skeleton annotations without the need to be trained for a specific input topology.
- we show how our method can be used to complement existing HPE approaches.

2. RELATED WORK

Human skeleton models. In HPE, the most common representation of the human body consists of a skeleton, namely a set of segments representing the limbs, with each limb connecting pairs of joints [6]. When annotating 2D datasets [7], the different joints are manually placed on the 2D image plane. As for 3D, reliable pose data [4, 9, 13, 15] cannot be obtained by manually annotating images taken in-the-wild.

Dataset	Ref.	Ground truth format	Number of joints
Human3.6	[9]	3D joint locations	17
Panoptic	[13]	3D joint locations	19
Panoptic2.0	[14]	3D joint locations	25
Panoptic2.0	[14]	SMPL (pseudo GT)	24
CMU MoCap	[4]	SMPL	24

Table 1: Different datasets are annotated according to different formats, either simple 3D skeletons or more complex skeleton-based 3D mesh models (SMPL).

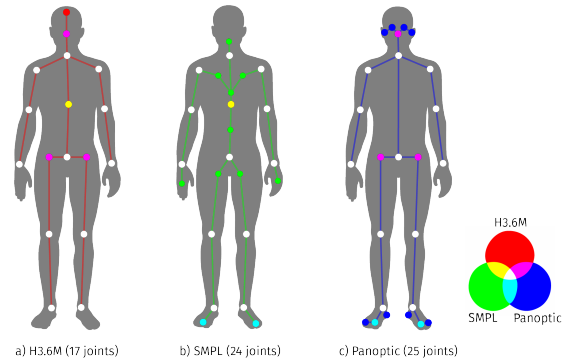


Fig. 2: Three different topologies used in common datasets: (a) H3.6M [9], (b) SMPL [4] and (c) Panoptic [13]. The primary RGB colors are used to identify the joints belonging to a single topology, white is used to highlight the joints in common across all 3 topologies, and the secondary colors are for the overlaps between pairs of topologies.

3D human pose datasets are usually acquired using motion capture systems [4], multiple cameras in controlled scenarios [9, 13] and synthetic environments [15].

Among the most commonly used datasets nowadays, Human3.6 [9] has enabled deep learning algorithms to be extensively validated. Other datasets, such as Panoptic [13], have also been largely adopted by researchers in the field [3], showing how to build proper 3D pose datasets without using motion capture systems. More recently, datasets [4, 15] based on the Skinned Multi-Person Linear model (SMPL) [8] have emerged and grown in popularity. The SMPL skeletal model is obtained through regression from the 3D body mesh, which allows for a better and more consistent 3D annotation compared to manual annotations.

Domain adaptation. When talking about skeletons, domain adaptation means translating the position or the topology of an existing skeleton into a different one. Generative Adversarial Models (GANs) have dealt with it by enforcing the discriminator to tell apart skeletons generated with a different topology from the original ones [16]. However, GAN-based methods are prone to mode-collapse, thus achieving poor generalization to different poses. Normalizing Flows (NFs) [2] have demonstrated the ability to map human pose distributions into a simpler representative subspace. Check-

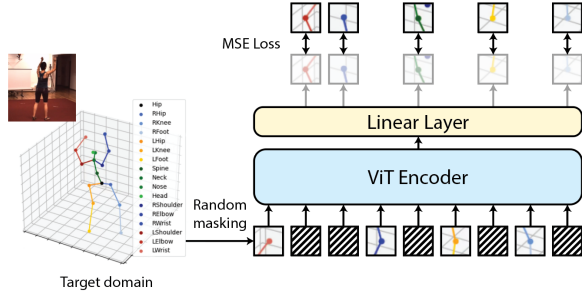


Fig. 3: Overview of our training strategy. Skeletons from the target domain are given as input to a Vision Transformer (ViT) encoder, with a subset M of randomly masked joints. A linear layer learns to reconstruct the masked joint at the output.

ing if the mapping of the sample in the new subspace belongs to the distribution has proven effective to discriminate new samples produced by a generative model. Inspired by [2], we employ NFs to obtain a likelihood metric of how well an adapted input skeleton fits the target domain distribution.

Masked modeling for representation learning. Masked language modeling [17] and masked image modeling [10, 11] have shown how self-supervised approaches can achieve better performances with respect to fully supervised ones, making them scalable representation learners for multiple tasks. Transformers architectures [12] have further contributed to improving representation learning when combined with a masking strategy on randomly selected image parts [1]. In particular, SimMIM [10] has proven to be a simple yet effective strategy, masking image patches by replacing them with random token vectors of the same dimension. In SimMIM, the goal is to train a network to obtain a strong image representation for the prediction task. Inspired by their work, we apply a similar strategy by randomly masking a subset of skeleton joints. We demonstrate the effectiveness of the self-supervised learned skeleton representation by using it in a skeletal topology domain adaptation task.

3. METHOD

Let $S_I = \{j_1, \dots, j_N\}$ be a representation of a human skeleton with N 3D joints $j_n = (j_x, j_y, j_z)$ in an input domain I . Our goal is to train a neural network able to adapt an input 3D skeleton S_I to a target 3D skeleton $\hat{S}_I = \{j'_1, \dots, j'_{N'}\}$ in a target domain T representing the same human pose varying either the position of joints $j_n \neq j'_{n'}$ or the total joints number $N \neq N'$, or both.

Embedding. Transformers usually process chunks of data as words in NLP [12] or image patches in computer vision [1]. Similarly to the work in [1], we choose individual joints as the data chunk to be embedded in a transformer architecture. The embedding is performed with a simple trainable linear layer, that transforms each joint into a token

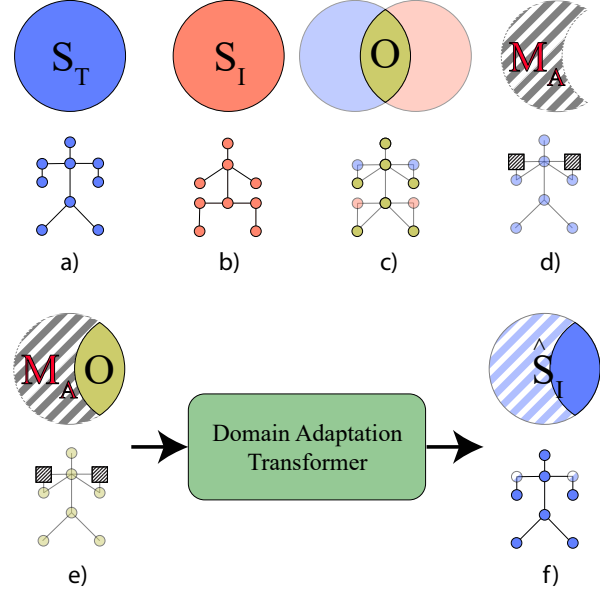


Fig. 4: Our domain adaptation procedure. A generic skeleton S_T in the target domain T (a) and a generic skeleton S_I in the input domain I (b) share the subset of common joints $O = S_T \cap S_I$ (c) and the subset of joints to be reconstructed is defined as $M_A = S_T \ominus O$ (d), which is to be masked during the domain adaptation. Thus, the shared subset O combined with the subset of masked joints to be reconstructed M_A (e) is the input to our DAT. The resulting adapted skeleton in the new topology domain T is $\hat{S}_I = O \oplus \widehat{M}_A$ (f), where $\widehat{M}_A$ are the joints predicted by the network.

vector of fixed dimensions d .

Domain Adaptation Transformer. In our architecture, called Domain Adaptation Transformer (DAT), the encoder has to extract a latent feature representation for the masked portion of the skeleton. The encoder consists of a tiny transformer [1]: we use such an architecture because of the token-based approach combined with the positional encoding. Compared to MLPs and CNNs, these two properties allow for a precise masking strategy of the feature vector representing a specific joint, while retaining the 3D space relationships. GANs, instead, are notably more unstable and difficult to train.

In our proposal, each skeleton is represented by N d -dimensional vectors which need to be passed to the encoding transformer [1] following a predefined order. This allows the architecture to learn the positional embedding, which represents the position of a joint in the embedding vector and its position in the 3D world relative to other joints. Instead of the prediction head (decoder) [1], we deploy a simple linear layer that has proven able to reconstruct the masked parts of the input [10]. Our transformer consists of 3.3M parameters only, compared to 86M of ViT-base or 632M of the ViT-huge.

Masking strategy. Given an arbitrary input skeleton, we

select a subset of joints $M = \{j_1, \dots, j_M\}$ to be masked. As in other works dealing with input masking [10, 17], we replace each masked joint j_m with a token vector of the same size d as the embedding. We investigate different masking strategies such as setting masked joints in the origin, in a random position within the skeleton boundary, or adding a random offset.

Training. Once a target domain T is chosen, we train our architecture to represent the target topology, by learning to reconstruct the masked part M of an input skeleton S_T , as shown in Figure 3. At training time, for each batch of samples we randomly select the set of joints M to be masked. The same M joints are selected for all the samples in the batch. Randomly selecting the M joints for each batch allows the network to learn a representation robust to different perturbations. The network is trained combining the Mean Square Error (MSE) loss and a bone length prior [2]. The total training loss is defined as:

$$\mathcal{L} = \frac{1}{M} \sum_{m=1}^M (j_m - \hat{j}_m)^2 + \mathcal{L}_{bone} \quad (1)$$

where j_m is a joint in M , $\hat{j}_m$ is a joint reconstructed by the network and $\mathcal{L}_{bone}$ is defined as in [2]. The masked pre-training is used in many different works to obtain a meaningful representation of the data [10, 11, 17], and it can be used for different down-stream tasks such as pose estimation [3].

Domain adaptation. As shown in Figure 2, different skeleton topologies share a subset of joints (e.g. pelvis, ankles, wrists) and differ in others (e.g. hips, shoulders, spine). The goal of the domain adaptation strategy is to transform S_I from an input domain I to $\hat{S}_I$, which represents the same body pose in the target domain T . While at training time we randomly select the M joints to be masked, we now need to properly select the M_A joints to be reconstructed, namely the ones missing in the skeleton S_I , to obtain the new skeleton $\hat{S}_I$. The procedure is described in Figure 4.

4. EXPERIMENTS

Datasets and topologies. To demonstrate the effectiveness of our approach, we test our DAT architecture on 3 datasets: (I) **H3.6M**, the Human3.6 Dataset [9] is annotated with a topology with $N = 17$ joints; (II) **CMU**, the CMU Graphics Lab Motion Capture Database [4] is annotated with a SMPL model, with $N = 24$ joints; (III) **PANO**, the Panoptic 2.0 [14] dataset is annotated with using $N = 25$ joints. It also provides a pseudo ground truth, named Pano(SMPL), in the SMPL format with $N = 24$. The topologies are shown in Figure 2.

Experimental setup and evaluation metrics. In the experiments, each dataset can arbitrarily be used as the input domain I and/or target domain T . The adaptation from an input to a target domain is denoted as $I \rightarrow T$.

To quantitatively evaluate the results using the standard Mean Per Joint Position Error (MPJPE) [3, 18], one would

need the ground truth data in both the input I and target T domains. In other applications where the ground truth of the target domain is not available, like in image super-resolution [19], a common approach is to downscale an image and up-scale it back to the original resolution, to evaluate the reconstructed image. Similarly, we deploy a double domain adaptation $I \rightarrow T \rightarrow \hat{I}$ to obtain a result comparable to the input domain. We can then compute the mean MPJPE between I and $\hat{I}$. If the ground truth is not available, the MPJPE is not applicable, thus we introduce a novel metric to evaluate the quality of the reconstructed skeleton without the ground truth.

Pose likelihood score. Normalizing Flows have proven effective in verifying if a skeleton belongs to a given domain distribution [2]. Similar to [2], we first project 3D poses to a lower dimensional space using a Principal Component Analysis (PCA) on which we train a NF of the input domain distribution I . Each block of the NF consists of two linear layers of size d_{NF} , connected by a ReLU. We feed each skeleton S through the NF to obtain a score $NF(S)$, which measures the likelihood that a skeleton belongs to the distribution of the input domain I . The pose likelihood score of a skeleton S to belong to domain I is defined as

$$P_L(S|I) = \frac{NF(S)}{\max(NF(I))} \% \quad (2)$$

where $\max(NF(I))$ is the highest value $NF(S_I)$ of likelihood scored by a skeleton S_I in the training set.

Implementation details. Our network is trained using PyTorch Lightning on a single NVIDIA GeForce RTX 3090. The token vector representing each joint has size $d = 128$. During training, we set the following hyper-parameters: 25 epochs, learning rate $2e^{-4}$, exponential LR scheduler with $\gamma = 0.95$, batch size 32, and Adam optimiser with weight decay $5e^{-2}$. The NF consists of 8 blocks, with each linear layer of size $d_{NF} = 1024$. The basis of the PCA reduction is set at [39, 55, 57] for the three datasets. During the training on the training set of each dataset, we set the following hyper-parameters: 25 epochs, learning rate $2e^{-4}$, a multi-step scheduler where the learning rate is divided by 10 at epochs 15 and 20, batch size 256, and a weight decay $1e^{-5}$.

4.1. Quantitative results

We report the quantitative results for each input domain I and target domain T pairs in Table 2. We report the mean μ and standard deviation σ of the pose likelihood score P_L computed on the test set of I given the distribution of each training set of I .

Single adaptation. In this experiment, we need the ground truth data of the same pose in both I and T , which is available only for the Panoptic 2.0 dataset [14], in the SMPL and Panoptic format. The results are reported in Table 3. We compare against a baseline, where the conversion from one

	I	T	M	$\hat{M}_A$	O	M_A	MPJPE (mm)	$P_L(\hat{I} I)$ (%) μ σ	
I	H3.6M	H3.6M	6					86.1	6.8
II	H3.6M	CMU	6	4			7.62	85.8	6.6
III	H3.6M	PANO	6	2			6.32	83.6	6.8
IV	CMU	CMU	13					85.9	6.5
V	CMU	H3.6M	13	11			9.36	62.9	15.2
VI	CMU	PANO	13	10			6.95	65.4	12.9
VII	PANO	PANO	14					89.0	5.5
VIII	PANO	H3.6M	14	10			11.24	77.5	4.9
IX	PANO	CMU	14	11			9.91	78.7	6.0

Table 2: Double adaptation results. For each $I \rightarrow T$, we perform the $I \rightarrow T \rightarrow \hat{I}$ adaptation. M is the number of joints masked while training DAT on T . $\hat{M}_A$ is the number of joints to be reconstructed while performing the domain adaptation. O and M_A identify the joints overlapped and to be masked, respectively, according to the color code in Fig. 2. The mean MPJPE is computed between I and $\hat{I}$, so as the likelihood score $P_L(\hat{I}|I)$, for which we report the mean μ and the variance σ . For experiments (I,IV,VII), $\hat{I}$ corresponds to the training set of I .

skeleton format to the other one is done following geometrical hand crafted rules similarly to [5]. The low MPJPE value is obtained going in both directions: adapting the SMPL ground-truth to the domain of the original Panoptic data and vice-versa. This shows how our DAT is able to effectively adapt an input domain to a target domain both changing the topology and number of joints, improving on the geometrical baseline when the target skeleton has either more and fewer joints than the input one.

Double adaptation. To demonstrate the effectiveness of our DAT when the ground truth of both domains is not available, we deploy the following cyclic transformation $I \rightarrow T \rightarrow \hat{I}$, thus obtaining the input domain $\hat{I}$, which corresponds to a first adaptation to T , and back to the original format of the input domain. Since I and $\hat{I}$ share the same topology, we can compute both metrics, showing how close the double adapted data $\hat{I}$ is to the input data I . This allows to implicitly validating also the intermediate target domain T . The results are reported in Table 2. Experiments (I,IV,VII) express the likelihood of a pose from the test set to belong to the same distribution of the training set (the higher, the better). The remaining experiments refer to the likelihood of double adapted test set $\hat{I}$ to belong to the input distribution of the training set of I (the higher, the better). The mean MPJPE expresses the difference between the original test set of I and $\hat{I}$. The double domain adaptation leads to results close to the original test (I) for the H3.6M dataset (II,III), since it is the one with fewer joints and thus less information to recover. Experiments on the CMU dataset (V,VI) show more pronounced differences from the input test set (IV). This dataset contains the most out-of-distribution poses, which are more

Exp	I	T	MPJPE (mm)	
			Ours	Baseline
I	PANO	PANO (SMPL)	1.10	4.05
II	PANO (SMPL)	PANO	2.00	7.20

Table 3: Single adaptation results obtained performing the $I \rightarrow T$ adaptation.

Exp	M	Strategy	P.E.	$P_L(\hat{I} I)$ (%)	
				μ	σ
I	1	zero	✓	79.56	6.3
II	3	zero	✓	79.16	6.4
III	6	zero	✓	78.87	8.8
IV	12	zero	✓	79.53	8.2
V	leaf	zero	✓	38.47	15.7
VI	6	random	✓	71.32	21.2
VII	6	noise	✓	71.43	21.2
VIII	6	zero	✗	18.07	18.3

Table 4: Ablation study results of different DAT configurations training on H3.6M. $\hat{I}$ is the test set of H3.6M; P.E. is the Positional Encoding.

difficult to be transformed to different domains. Experiments on the Panoptic2.0 dataset (VIII,IX) return results that are closer to the input test set (VII). This happens because of the size of the dataset which provides the NF with a huge variety of poses with a high likelihood score.

Ablation studies. We perform the ablation studies on various training configurations of H3.6M, reporting the $P_L(S|I)$ for all the generated skeletons S in the test set. The results are reported in Table 4. Experiments (I,II,III,IV) study the variation of the number of masked joints M between 1 and 12 at training time on the test set. DAT is able to correctly retrieve the test data distribution regardless of how many joints are masked; as expected, the uncertainty increases as the number of masked joints increases. (III) is chosen as the baseline for our experiments because we need to mask a number of joints $M > M_A$. In (V) we masked all the joints but the leaf joints (end effectors of the kinematic chain), which leads to a decrease of both μ and σ . Experiments (III,VI,VII) investigate different masking strategies, such as setting all the nodes to the axis origin (III), or randomly in the 3D normalised space (IV) or adding random noise (VII), with masking strategy in (III) outperforming the others. In (VIII), we remove the positional encoding (P.E.), leading to much worse results, showing its key role in modeling the relationships between joints.

4.2. Qualitative results

In Figure 5 we show some qualitative results obtained adapting skeletons from an input domain I to a target domain T . We train and test on all the possible $I \rightarrow T$ combinations of the 3 datasets. Adapting a representation with a higher

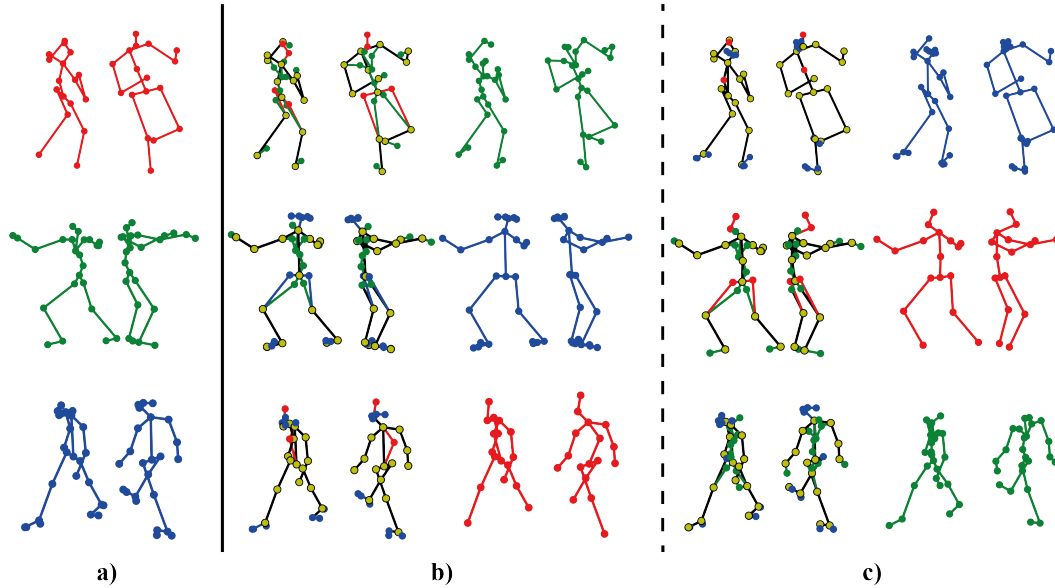


Fig. 5: Qualitative results. Legend: **H3.6M**, **SMPL**, **Panoptic**. (a) We show the frontal and side view of an input domain skeleton I to be transformed to the two other target domains T . (b) and (c) show the transformation of each skeleton from the input domain I to a different target domain T . The first skeleton represents each skeleton from I superimposed with the transformed version in T with the common joints highlighted in yellow. The second skeleton represents the transformed skeleton in the target domain T .

number of joints to another with a lower one (e.g. **SMPL** $\rightarrow$ **H3.6M**, removing joints along the spine) is the simplest scenario, since some joints are not given as an input to the network. It is interesting to notice how the domain adaptation network works when adding novel joints in 3 different cases of increasing complexity:

1. adding joints subdividing existing bones of the original topology (e.g. adding a novel joint along the spine **H3.6M** $\rightarrow$ **SMPL**)
2. adding joints to change the topology of a selected body part (e.g. changing the hips topology from angled to flat and vice-versa **H3.6M** $\leftrightarrow$ **Panoptic**)
3. adding joints falling outside the original skeleton boundaries (e.g. adding novel joints for the feet or the ears and eyes **H3.6M** $\rightarrow$ **Panoptic**)

In the third case, the network correctly guesses the position and orientation of novel joints (e.g. feet and ears/eyes). This is particularly remarkable, because the network needs to have a deeper understanding of the overall pose in order to make a correct guess.

The real example in Fig. 6, shows how our approach can help existing method in reconstructing the position of masked 3D joints that are usually missing or predicted with low confidence.

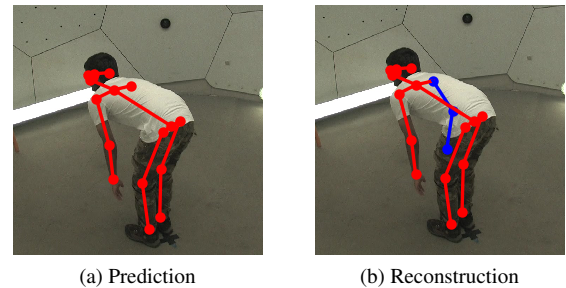


Fig. 6: Classic pose predictors (a) sometimes fail to detect some joints or entire limbs due to strong occlusions or out-of-image coordinates. As occlusions can be interpreted as masked joints, our method (b) can reconstruct an arbitrary number of joints in the incomplete pose even for imperfect predictions.

5. CONCLUSIONS

We present a novel Domain Adaptation Transformer (DAT) to solve the human pose domain adaptation problem. We believe that providing researchers with a method to adapt the ground truth across datasets could help them focus more on developing HPE solutions, without struggling to incorporate different annotation formats. In this work, we show how a self-supervised masking strategy effectively helps the network to learn a representation of the human pose, which we use to adapt skeletons from multiple different domains.

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